Lab Viva

Basic Questions About Latch

Q1. What is a latch?
Ans: A latch is a basic memory circuit that stores one bit of information (0 or 1).

Q2. How many inputs and outputs does a basic latch have?
Ans: It has two inputs and two outputs (Q and $\overline{Q}$).

Q3. What is the main function of a latch?
Ans: Its main function is to store binary information.

Q4. Is a latch a sequential circuit?
Ans: Yes, because its output depends on both present inputs and previous states.

Q5. Is a latch level-triggered or edge-triggered?
Ans: A latch is level-triggered.

Q6. What is the difference between a latch and a flip-flop?

Latch	**Flip-Flop**
Level-triggered	Edge-triggered
No clock is required in a basic latch	Requires a clock
Simpler circuit	More complex circuit
Stores 1 bit	Stores 1 bit

Q25. What is a clock signal?
Ans: A clock signal is a periodic signal used to control when a flip-flop changes its state.

Q26. What is edge triggering?
Ans: Edge triggering means the flip-flop changes its output only at the rising or falling edge of the clock.

Q27. What is the difference between positive and negative edge triggering?
Ans:

- Positive edge: Output changes at the rising edge (0 to 1).
- Negative edge: Output changes at the falling edge (1 to 0).

Q28. What is a JK flip-flop?
Ans: A JK flip-flop is an improved SR flip-flop that has no invalid input condition.

Q29. What happens when J = 1 and K = 1?
Ans: The output toggles (changes from 0 to 1 or 1 to 0).

Q30. What is a D flip-flop?
Ans: A D flip-flop stores the value of its D input at the active clock edge.

Q31. What is a T flip-flop?
Ans: A T flip-flop changes its output when T = 1 and holds its previous state when T = 0.

Q32. What is race-around condition?
Ans: Race-around occurs in a level-triggered JK flip-flop when J = K = 1 and the clock pulse is too long, causing repeated output toggling.

Q33. How can race-around condition be eliminated?
Ans:

1. Using Master-Slave JK flip-flop.
2. Using Edge-triggered flip-flop.
3. Using a narrow clock pulse.

Invalid State: An invalid state is a condition in a latch or flip-flop where the outputs do not maintain their required complementary relationship ($Q \neq \overline{Q}$).

D Flip-Flop

Q1. What is a D flip-flop?
Ans: A D flip-flop is a memory circuit that stores one bit of data.

Q2. Why is it called a D flip-flop?
Ans: D stands for Data or Delay.

Q3. How many inputs does a D flip-flop have?
Ans: One main input (D) and one clock input.

Q4. What is the characteristic equation of a D flip-flop?
Ans: Q(next) = D.

Q5. Does a D flip-flop have an invalid state?
Ans: No, it has no invalid input condition.

T Flip-Flop

Q6. What is a T flip-flop?
Ans: A T flip-flop is a circuit that toggles its output when T = 1.

Q7. What does T stand for?
Ans: T stands for Toggle.

Q8. What happens when T = 0?
Ans: The output holds its previous state.

Q9. What happens when T = 1?
Ans: The output toggles (0 becomes 1 and 1 becomes 0).

Q10. What is the characteristic equation of a T flip-flop?
Ans: $Q(next) = T \oplus Q$.

Important Truth Tables

D Flip-Flop

D	Q(next)
0	0
1	1

T Flip-Flop

T	Q(next)	Operation
0	Q	Hold
1	$\overline{Q}$	Toggle

Bonus: Difference Between D and T Flip-Flop

D Flip-Flop	T Flip-Flop
Data flip-flop	Toggle flip-flop
One input (D)	One input (T)
Q(next) = D	Q(next) = T ⊕ Q
Stores input data	Toggles output
Used in registers	Used in counters

Viva Tip: Remember these two equations:

- D Flip-Flop: **Q(next) = D**
- T Flip-Flop: **Q(next) = T ⊕ Q**

1. Encoder

Q1. What is an encoder?
Ans: An encoder is a combinational circuit that converts 2^n input lines into n output lines.

Q2. How many output lines does an encoder have?
Ans: For 2^n input lines, it has n output lines.

Q3. What is an 8-to-3 encoder?
Ans: It has 8 input lines and 3 output lines.

Q4. What is the main function of an encoder?
Ans: It converts input information into binary code.

Q5. Is an encoder a sequential or combinational circuit?
Ans: Combinational circuit.

Q6. What happens if two inputs are HIGH simultaneously in a basic encoder?
Ans: The output may become ambiguous or incorrect.

Q7. Give one application of an encoder.
Ans: Keyboard encoding and digital communication.

Q8. What is a 4-to-2 encoder?
Ans: It converts 4 input lines into 2 binary output lines.

2. Priority Encoder

Q9. What is a priority encoder?
Ans: A priority encoder is an encoder that gives priority to the highest-priority active input.

Q10. Why do we use a priority encoder?
Ans: To handle multiple active inputs without ambiguity.

Q11. What happens when multiple inputs are HIGH in a priority encoder?
Ans: Only the highest-priority input is encoded.

Q12. What is the difference between an encoder and a priority encoder?

Encoder	Priority Encoder
Normally assumes only one input is active.	Allows multiple active inputs.
No priority system.	Has a priority system.
Multiple active inputs cause ambiguity.	Selects the highest-priority input.

Q13. What is the highest-priority input in an 8-to-3 priority encoder?
Ans: Usually I7, depending on the design.

Q14. What is a valid bit in a priority encoder?
Ans: It indicates whether at least one input is active.

Q15. Give one application of a priority encoder.
Ans: Interrupt handling in microprocessors.

1. Multiplexer (MUX)

Q1. What is a multiplexer (MUX)?
Ans: A multiplexer is a combinational circuit that selects one input from multiple inputs and sends it to a single output.

Q2. How many select lines are required for a 4:1 MUX?
Ans: 2 select lines.

Q3. How many inputs and outputs does an 8:1 MUX have?
Ans: 8 inputs and 1 output.

Q4. What is the main function of a MUX?
Ans: To select one input and transfer it to the output.

Q5. Is MUX a combinational or sequential circuit?
Ans: Combinational circuit.

2. Demultiplexer (DEMUX)

Q6. What is a demultiplexer (DEMUX)?
Ans: A demultiplexer is a combinational circuit that takes one input and sends it to one of several outputs.

Q7. How many select lines are required for a 1:8 DEMUX?
Ans: 3 select lines.

Q8. How many inputs and outputs does a 1:4 DEMUX have?
Ans: 1 input and 4 outputs.

Q9. What is the main function of a DEMUX?
Ans: To distribute one input to a selected output.

Q10. What is the difference between MUX and DEMUX?

MUX	DEMUX
Many inputs, one output	One input, many outputs
Data selector	Data distributor
4:1, 8:1	1:4, 1:8
Parallel to serial selection	Serial to parallel distribution

Q1. What is a decoder?
Ans: A decoder is a combinational circuit that converts n input lines into a maximum of 2^n output lines.

Q2. How many outputs does a 3-to-8 decoder have?
Ans: 8 outputs.

Q3. How many input lines does a 2-to-4 decoder have?
Ans: 2 input lines.

Q4. What is the main function of a decoder?
Ans: To convert binary input into a specific output signal.

Q5. What is the difference between encoder and decoder?

Encoder	Decoder
Converts input into binary code.	Converts binary code into output signals.
2^n inputs, n outputs.	n inputs, 2^n outputs.

1. MOD Counter

Q1. What is a MOD counter?
Ans: A MOD counter is a counter that counts a fixed number of states before repeating its sequence.

Q2. What does MOD-8 mean?
Ans: It means the counter has 8 different states (0 to 7).

Q3. How many flip-flops are required for a MOD-8 counter?
Ans: 3 flip-flops.

Q4. What is the formula for the maximum number of states?
Ans: MOD = 2^n, where n is the number of flip-flops.

Q5. How many flip-flops are required for a MOD-10 counter?
Ans: 4 flip-flops.

Q6. What is a truncated counter?
Ans: A counter that uses fewer states than its maximum possible 2^n states.

2. Asynchronous Counter

Q7. What is an asynchronous counter?
Ans: An asynchronous counter is a counter in which only the first flip-flop receives the external clock.

Q8. Why is it called an asynchronous counter?
Ans: Because all flip-flops do not receive the clock simultaneously.

Q9. What is the main disadvantage of an asynchronous counter?
Ans: Propagation delay increases with the number of flip-flops.

Q10. Which flip-flops are commonly used in asynchronous counters?
Ans: T flip-flops and JK flip-flops.

Q11. What is the clock connection in an asynchronous counter?
Ans: The output of one flip-flop is connected to the clock input of the next flip-flop.

3. Up-Down Counter

Q12. What is an up counter?
Ans: An up counter counts in increasing order, such as 0, 1, 2, 3.

Q13. What is a down counter?
Ans: A down counter counts in decreasing order, such as 3, 2, 1, 0.

Q14. What is an up-down counter?
Ans: A counter that can count both upward and downward.

Q15. What determines the counting direction of an up-down counter?
Ans: The control input determines the counting direction.

Q16. What is the main application of an up-down counter?
Ans: Digital clocks, elevators and counting systems.

4. Ripple Counter

Q17. What is a ripple counter?
Ans: A ripple counter is an asynchronous counter in which the output of one flip-flop triggers the next flip-flop.

Q18. Why is it called a ripple counter?
Ans: Because the clock effect propagates from one flip-flop to the next like a ripple.

Q19. What is the difference between a synchronous and asynchronous counter?

Synchronous Counter	Asynchronous Counter
All flip-flops receive the same clock.	Only the first flip-flop receives the external clock.

Synchronous Counter	Asynchronous Counter
Faster operation.	Slower operation due to accumulated delay.
Less ripple delay.	Has ripple delay.

Q20. What is propagation delay in a ripple counter?
Ans: It is the time required for the output change to propagate through all flip-flops.

Topic	Formula
Maximum states	2^n
Number of flip-flops	$n = \log_2(MOD)$
MOD-4	2 flip-flops
MOD-8	3 flip-flops
MOD-16	4 flip-flops
MOD-32	5 flip-flops

Where n = Number of flip-flops.

1. **Can a 3-bit counter count up to 8?**
   Ans: It has 8 states, from 0 to 7.
2. **Which counter is faster: synchronous or asynchronous?**
   Ans: Synchronous counter.
3. **Does a ripple counter require a common clock for all flip-flops?**
   Ans: No.
4. **What happens after the last state of a MOD-8 counter?**
   Ans: It returns to state 000.
5. **Can a MOD-10 counter be made using 4 flip-flops?**
   Ans: Yes, because 4 flip-flops provide 16 possible states, and 10 states are used.